

Game Theory-Assignment4

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Exercise 10

We compute the value and a behavioral saddle-point equilibrium of the extensive-form feedback game using recursive backward induction.

1. Behavioral Saddle-Point Equilibrium

Each multi-node information set is solved as a local zero-sum matrix game. Its saddle-point value replaces that information set.

Terminal Information Sets

Single-node information sets These are simple decision nodes of Player 2. The optimal action maximizes Player 2's payoff.

$$v_{\beta_2} = 1, \quad v_{\beta_3} = 1.$$

Information set β_4

$$A_{\beta_4} = \begin{array}{c|cc} & L & R \\ \hline \alpha_3 L & -2 & -1 \\ \alpha_3 R & 1 & -1 \end{array}$$

Row minima:

$$-2, -1$$

Column maxima:

$$1, -1$$

Saddle point:

$$(\alpha_3 R, R)$$

$$v_{\beta_4} = -1$$

Information set β_5

$$A_{\beta_5} = \begin{array}{c|cc} & L & R \\ \hline \alpha_4 L & -1 & -3 \\ \alpha_4 R & -3 & -1 \end{array}$$

No pure saddle point. Solve mixed equilibrium.

Let Player 1 choose L with probability p .

Equalize column payoffs:

$$-1p - 3(1 - p) = -3p - 1(1 - p)$$

$$-3 + 2p = -1 - 2p \Rightarrow p = \frac{1}{2}.$$

Both players mix uniformly.

$$v_{\beta_5} = -2$$

Information set β_6

$$A_{\beta_6} = \begin{array}{c|cc} & L & R \\ \hline \alpha_5 L & -1 & 2 \\ \alpha_5 R & -1 & 2 \end{array}$$

Column L minimizes Player 1 payoff.

$$v_{\beta_6} = -1$$

Reduced Game at β_1

Replace downstream values:

$$A_{\beta_1} = \begin{pmatrix} 1 & -1 \\ -2 & 2 \end{pmatrix}$$

Row minima:

$$-1, -2$$

Column maxima:

$$1, 2$$

Equilibrium payoff:

$$V = 0.$$

Behavioral Equilibrium Strategies

Player 1:

$$\alpha_1 : T, \quad \alpha_3 : R, \quad \alpha_4 : \text{mix } (0.5, 0.5).$$

Player 2:

$$\beta_1 : L, \quad \beta_4 : R, \quad \beta_5 : \text{mix } (0.5, 0.5), \quad \beta_6 : L.$$

Game Value

$$V = 0$$

2. Strategic-Form Verification (MATLAB)

MATLAB Implement

```
1 \begin{verbatim}
2 \begin{verbatim}
3 clc; clear;
4
5 A = [ 1  -1;
6      -2  2 ];
7
8 [m,n] = size(A);
9
10 f = [zeros(m,1); 1];
11
12 Aineq = [A', -ones(n,1)];
13 bineq = zeros(n,1);
14
15 Aeq = [ones(1,m), 0];
16 beq = 1;
17
18 lb = [zeros(m,1); -Inf];
19
20 x = linprog(f,Aineq,bineq,Aeq,beq,lb);
```

```

21
22 p = x(1:m);
23 v = x(end);
24
25 fprintf('Mixed value of the game: %.4f\n', v);
26 fprintf('Optimal mixed strategy:\n');
27 disp(p)

```

Listing 1: MATLAB code for Exercise 10

Output

$$V = 0$$

Matches behavioral solution.

3. Final Result

- Behavioral equilibrium obtained by backward induction.
- Strategic-form LP gives identical value.

$$\text{Game value} = 0$$

Homework #4**Student name :**

Exercise	weight	grade (0..4)
10.1	50%	3
10.2	50%	4

Grades

4 – correct

3 – mostly correct

2 – half-and-half

1 – mostly incorrect

0 – not done